

ATEK1601

Programming Supplement

2-18 GHz Sub-Octave USB-Controlled Filter Bank

Document revision 1.0 | Python API version 1.0.0

This supplement contains the ATEK1601-specific programming information. It is used together with the **ATEK RF Modules Programming Manual**, which describes the connection, the command set, state reporting and the Python API common to all modules.

Software, example programs and datasheets: <https://github.com/atek-midas/atek-rf-modules>

1 Module Identification

Item	Value
Module name (*IDN? response)	ATEK1601
RF chip (datasheet)	ATEK656N5
Description	2-18 GHz Sub-Octave USB-Controlled Filter Bank
States	6 (state 0 ... 5)
Python class	atek.ATEK1601

RF performance data is found in the datasheet of the RF chip.

2 States

6-band sub-octave filter bank. Band n is selected with state $n - 1$. The front-panel buttons step through the same states.

6 bands. Pass bands:

State	Band	Pass band (GHz)
0	Band 1	1.9 – 3.5
1	Band 2	2.8 – 5.4
2	Band 3	4.5 – 9.1
3	Band 4	7.1 – 12.3
4	Band 5	9.9 – 15.3
5	Band 6	12.5 – 18

3 Raw Command Example

Direction	Line	Comment
PC → Module	*IDN?	Identify the module
Module → PC	ATEK1601	
PC → Module	SET:3	Select band 4
Module → PC	STATE:3	Confirmation
PC → Module	GET?	Read the current state
Module → PC	STATE:3	Band 4, 7.1–12.3 GHz

Command syntax and state reporting are described in Section 3 of the ATEK RF Modules Programming Manual.

4 Python API

4.1 Class atek.ATEK1601

`atek.connect()` returns an `atek.ATEK1601` object when a ATEK1601 is connected. In addition to the common methods (ATEK RF Modules Programming Manual, Section 5.5) it provides:

Method / attribute	Description
<code>set_band(band, verify=True)</code>	Selects band 1...6. Returns the band.
<code>get_band()</code>	Returns the active band (1...6).
<code>get_band_range_mhz(band=None)</code>	Returns (low, high) of a band in MHz; default is the active band.
<code>set_frequency_mhz(mhz, verify=True)</code>	Selects the band that best covers mhz (the band in which mhz is closest to the band centre). Returns the band.
<code>num_bands</code>	Number of bands (6).
<code>BANDS_MHZ</code>	Tuple of the (low, high) pass bands in MHz, index = state.

Band numbers start at 1; state indexes start at 0. Frequencies are given in MHz.

4.2 Example program

```
import atek_rf_modules_scp_api as atek

with atek.connect("COM5") as fb:          # returns an atek.ATEK1601 object
    print("IDN      :", fb.identify())    # ATEK1601

    fb.set_band(3)                        # band 3 (state 2)
    print("Band    :", fb.get_band(), "->", fb.get_band_range_mhz(), "MHz")

    band = fb.set_frequency_mhz(10000)    # best band for 10 GHz
    print("10 GHz  :", "band", band, "->", fb.get_band_range_mhz(), "MHz")
```

Output:

```
IDN      : ATEK1601
Band     : 3 -> (4500, 9100) MHz
10 GHz   : band 4 -> (7100, 12300) MHz
```